

Extraction of β -carotene from peppers by means of certain organic solvents.

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Abstract. The aim of this paper is to present a study of certain characteristics of sweet peppers (genus *Capsicum* and species *Annuum*), grown in a private garden located in the North-Eastern region of the country (Razgrad District). 12 types of sown pepper seeds are considered. 11 types of them have annually been collected from the best quality fruits and selected by an amateur gardener. Only one group of pepper seeds (number 7) has been purchased commercially. Before conducting the experiment, the collected peppers have been stored in a refrigerator (5 °C) within a week. The following characteristics of the peppers were studied: moisture content, dry matter and ash content. Static extraction of β -carotene from the peppers was carried out in four organic solvents, used as extractants (isopropanol, isobutanol, 1,1-dichloromethane, 1,2-dichloroethane). The different organic solvents exhibited different extraction ability. The largest amount of β -carotene was extracted when polar organic solvents were used as extractants and first in terms of extraction ability was isopropanol, followed by isobutanol. The best organic solvent for extracting β -carotene from peppers proved to be isopropanol. The highest amount of β -carotene extracted by using isopropanol, was contained in sample № 7 (11.86 ± 0.01 mg/100g of dry matter).

1 Introduction

Carotenoids (from Latin "Carota" - carrot and from Greek "Eidos" - species) are natural yellow to red-orange pigments, synthesized by bacteria, algae, fungi, higher plants, corals and other organisms. They determine the color of flowers and fruits [1].

Carotenoids are a group of widely spread in nature organic compounds - organic pigments, found in the cells of plants and microorganisms and having a strong antioxidant effect. Some of them are biologically active substances. They are contained in high doses in a number of edible plant products, but the human body cannot synthesize them. This calls for their introduction from the outside, and the most convenient way for this is precisely through regular consumption of foods, rich in carotenoids [2, 3].

Beta-carotene is the best-known vitamin of the carotenoid group, present in fruits and vegetables. Of the approximately 50 types of carotenoids included in the human diet, β -carotene is the most common, followed by alpha-carotene, beta-cryptoxanthin, lutein, zeaxanthin, and lycopene [4].

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As industry and economy developed, and as living standards rose, dietary changes occurred. People today are increasingly turning their attention to new ways of feeding, and the modern knowledge of food physiology determines the growing importance of fruits and vegetables [5].

Vegetables are naturally rich in carbohydrates, organic acids, phytoncides, biologically active substances (vitamins, hormones, enzymes), basic macronutrients, micronutrients, amino acids and other substances, necessary for normal human growth, for development and normal functioning of the physiological and metabolic processes in the human organism [6]. Vegetables are an important source of proteins, vitamin A, vitamin C and minerals such as potassium, sodium and calcium [7].

Pepper is the second most important vegetable crop in the world, after tomato [8]. The brightness of its color is due to the presence of carotenoids [9]. β -Carotene is known as provitamin A and is a major source of vitamin A for humans. Thus, products, containing β -carotene, are of great importance for people. The presence of vitamin A in pepper is due to the high content of carotene (β -carotene) and cryptoxanthin. It has also been found that ripe pepper fruits contain almost the same amount of vitamin A as carrots [10,11]. In addition, red peppers are a good source of tocopherols, particularly α -tocopherol (vitamin E).

The most common way of obtaining carotenoids is their extraction from plant material, which is based on physicochemical processes. The carotenoids are soluble in organic solvents but insoluble in water (except crocin). Many of them dissolve in ethanol, petroleum ether, chloroform, and very well in fats [12]. The extraction proceeds better in polar organic solvents than in non-polar ones [13].

The aim of this paper is to determine some characteristics of sweet peppers, to conduct a static extraction of β -carotene from them with the help of four organic solvents used as extractants (isopropanol, isobutanol, 1,1-dichloromethane, 1,2-dichloroethane) and to quantify the β -carotene content of the obtained extracts.

2 Materials and methods

2.1 Materials

Sweet peppers (*Capsicum* genus and *Annuum* species) were used for the experiment. Early spring sowing of 12 types of pepper seeds was carried out in March 2018 in a private garden in the North-East region of the country (Razgrad district). Three groups of seeds were used:

- collected annually from the best quality fruits and selected by an amateur gardener for more than 40 years – the peppers with numbers 1, 2, 3, 5, 6, 8, 9 and 12;
- collected annually from the fruits of the best quality and selected by an amateur gardener for more than 10 years – the peppers with numbers 4, 10 and 11;
- purchased from the commercial network in 2018 – the peppers number 7.

The peppers were harvested from the garden at the beginning of September 2018, at the moment of their technological maturity and were characterized by an intense color for the respective species (Fig. 1). Before conducting the experiment, they were stored for one week in a refrigerator (5°C). The following reagents were used to conduct the experiments: isopropanol (Reachem, Slovak Republic), isobutanol (Reachem, Slovak Republic), 1,1-dichloromethane (Reachem, Slovak Republic), 1,2-dichloroethane (Reachem, Slovak Republic).



Fig. 1. Appearance of the studied peppers.

2.2 Determination of moisture content, mineral substances (ash content) and dry matter (dm) in the peppers

The moisture content (W , %) is determined according to the ISO 662:2016 method [14]. The amount of mineral substances (ash content) was determined according to the ISO 763:2003 method [15]. The quantity of dry matter is calculated by the formula (1):

$$DM = 100 - W, \% \quad (1)$$

2.3 Extraction of β -carotene from peppers

The extraction of β -carotene from the different average test samples of the peppers is carried out using 4 organic solvents: 2 polar type - isopropanol and isobutanol and 2 non-polar ones - 1,1-dichloromethane, 1,2-dichloroethane.

5g of the average test sample is placed in 150 cm³ beakers. 20.0g of the corresponding solvent is added to each sample. The static extraction is carried out at room temperature for 60 minutes. The samples are stirred every 10 minutes. After the time has elapsed, each sample is filtered through a paper filter. The obtained extracts are subjected to analysis.

2.4 Determination of the β -carotene content in the extracts

The methodology, presented by Fikselová, Šilhár, Mareček, & Frančáková, 2008 [16] was used to determine the amount of β -carotene in the extracts. In a separatory funnel, 20 cm³ of petroleum ether are pipetted. Distilled water is added to the resulting mixture to separate the

phases. The petroleum ether phase with the extracted β -carotene in it is placed in a 50 cm³ volumetric flask. The volume in the flask is brought up to 50 cm³ with petroleum ether. The obtained samples are filtered through a paper filter. The absorption of the resultant filtrates is measured using a UV-VIS spectrophotometer at a wavelength of 450 nm. The amount of β -carotene is determined by means of a standard curve for β -carotene in $\mu\text{g}/\text{cm}^3$. Three measurements are made for each sample. The obtained results are presented in mg/100 g of dry matter.

2.5 Statistical analysis

Standard deviation and Pearson correlation coefficient was performed with the software’s XLSTAT 2019 and Microsoft Office Excel 365.

3 Results and discussion

Pepper color brightness is due to the presence of carotenoids [17]. β -Carotene is known as provitamin A, since it is a major source of vitamin A for people. Vitamin A is necessary for normal human development, growth and vision. Thus, products, containing β -carotene, are essential for humans [18]. In addition, red peppers are a good source of tocopherols, particularly α -tocopherol (vitamin E) [19]. Figure 2 shows the dry matter and the moisture content of the studied peppers.

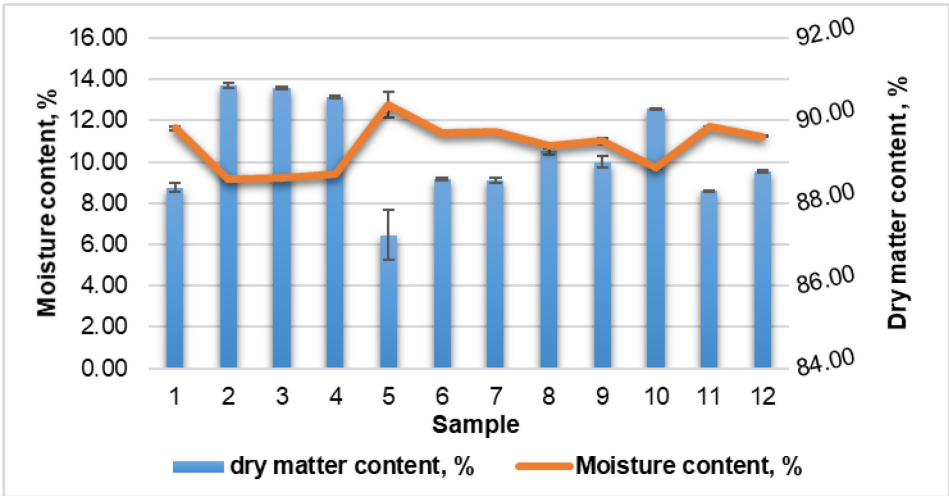


Fig. 2. Moisture content and dry matter of the peppers.

The average moisture content in the studied peppers was 10.79±0.10%. The highest moisture content and the lowest dry matter was found in the peppers from sample №1 (11.62±0.11% and 88.38±0.11%, respectively). On the other hand, the lowest moisture content and the highest dry matter was determined in peppers from sample №2 (9.16±0.06% and 90.84±0.06%, respectively).

The ash content in the peppers is presented in Figure 3. From the results can be seen that the average amount of mineral substances contained in the tested pepper samples is 0.98±0.02%. The highest ash content was determined in sample № 3 (1.74±0.02%), and the lowest - in sample № 10 (0.23±0.04%). Figures 4 show the content of β -carotene in the studied peppers, extracted in the different organic solvents. The values shown are the arithmetic mean of three consecutive tests.

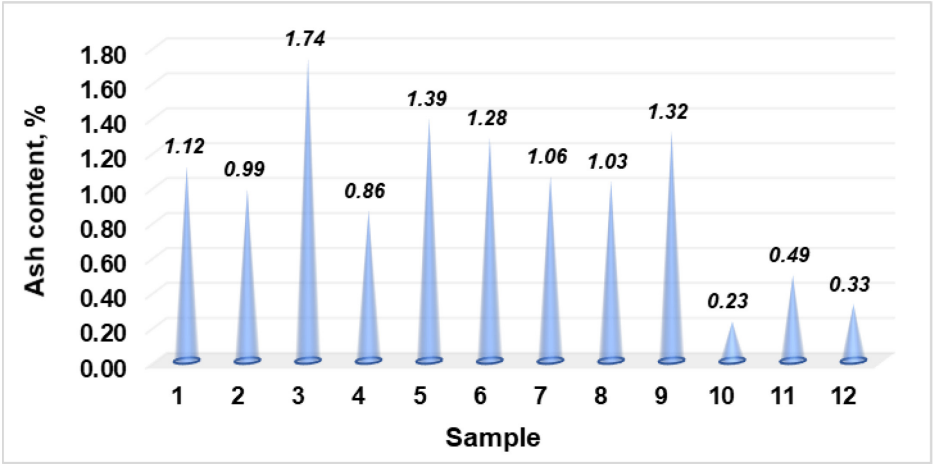


Fig. 3. Ash content in the peppers.

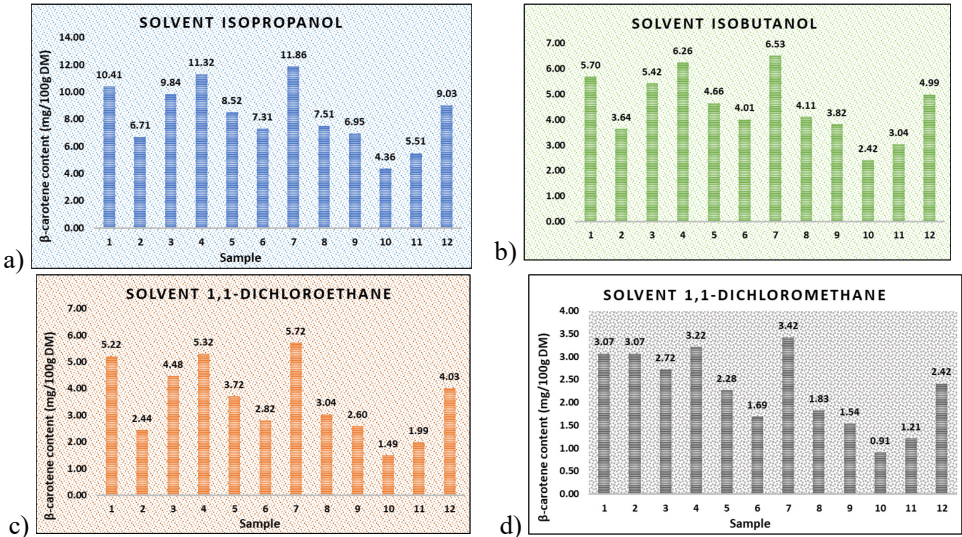


Fig. 4. Content of β-carotene in the peppers, extracted in: a) isopropanol; b) isobutanol; c) 1,1-dichloroethane; d) 1,1-dichloromethane.

From the presented results, it can be seen that the amount of β-carotene in the studied red peppers is in the range from 0.90 ± 0.01 to 11.87 ± 0.01 mg/100g of dry matter. The maximum β-carotene content extracted with isopropanol (fig. 4a) was 11.86 ± 0.01 mg/100g of dry matter, and the minimum was 4.36 ± 0.01 mg/100g of dry matter. When isobutanol (fig. 4b) was used as a solvent, the maximum amount of β-carotene was 6.53 ± 0.01 mg/100g of dry matter. The minimum amount of β-carotene in the investigated red peppers, extracted using isobutanol, was 2.42 ± 0.02 mg/100g of dry matter. When using 1,2-dichloroethane as a solvent (fig. 4c), the maximum amount of β-carotene contained in the studied peppers was 5.72 ± 0.01 mg/100g of dry matter, and the minimum was 1.49 ± 0.01 mg/100g of dry matter. The maximum content of β-carotene in the peppers, extracted with 1,1-dichloromethane (fig. 4d), was 3.42 ± 0.03 mg/100g of dry matter, while the minimum amount of extracted β-carotene was 0.91 ± 0.01 mg/100g of dry matter.

The presented results also show that when isopropanol is used as a solvent, the amounts of β -carotene are higher, compared to those, when using the other solvents. On the other hand, if the individual pepper samples are compared to one another, it can be seen that the largest amount of β -carotene is contained in the extracts of sample №7 (11.87 ± 0.01 mg/100g of dry matter, extracted with isopropanol; 6.53 ± 0.01 mg/100g of dry matter extracted with isobutanol; 5.72 ± 0.01 mg/100g of dry matter extracted with 1,2-dichloroethane and 3.42 ± 0.01 mg/100g of dry matter extracted with 1,1-dichloromethane).

Table 1. Pearson correlation coefficient

	Isopropanol	Isobutanol	1,1-dichloroethane	1,1-dichloromethane	Dry matter	Moisture	Ash
Isopropanol	1						
Isobutanol	0.999834	1					
1,1-dichloroethane	0.992373	0.992646	1				
1,1-dichloromethane	0.992813	0.993325	0.999471	1			
Dry matter	-0.0594	-0.05594	-0.0823	-0.07877	1		
Moisture	0.059397	0.05594	0.082303	0.078773	-1	1	
Ash	0.397239	0.388402	0.347425	0.35077	-0.04969	0.049692	1

The correlation between quantity of β -carotene extracted with different solvent, dry mater and ash in peppers are presented in table 1. From the results can see that quantity of β -carotene extracted with different solvent have very strong correlation between all used solvent, and weak correlation with quantity of ash in peppers. Extremely negative correlation was find between dry matter and moisture in peppers.

4 Conclusion

The amount of β -carotene extracted from 12 types of peppers by means of four organic solvents used as extractants (isopropanol, isobutanol, 1,1-dichloromethane, 1,2-dichloroethane) has been studied and reported in this paper. It has been proved that the different organic solvents exhibit different extraction ability. The largest amount of β -carotene is extracted when polar organic solvents are used as extractants and isopropanol proved to have the highest extraction ability, followed by isobutanol. The highest amount of β -carotene extracted with isopropanol was contained in sample № 7 (11.86 ± 0.01 mg/100g of dry matter).

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